The great tit HapMap project: a continental-scale analysis of genomic variation in a songbird

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Abstract

A major aim of evolutionary biology is to understand why patterns of genetic variation vary within genomes, and among populations and species. Widespread continental species are interesting models for studying how evolution shapes patterns of genomic divergence, as they are associated with recent and rapid colonisations, large population sizes and high levels of gene flow. We studied patterns of genomic variation among 28 great tit (*Parus major*) populations in Europe. We found that genome-wide variation was consistent with a recent colonisation across Europe from a single refugium in the Balkans and/or Turkey. The highest levels of genome-wide structure were found between island populations in Crete, Sardinia and Corsica, most likely as a result of bottlenecks and reduced gene flow among these populations. We then found that the landscape of genomic differentiation was highly heterogeneous, with clear "islands of differentiation", even among the most recently separated populations. These islands of differentiation were consistently found in regions of low recombination, suggesting that background selection can rapidly promote population differentiation among recently colonised populations. We also detected outlier regions that were unique to peripheral great tit populations, most likely as a result of recent natural selection at the range edges of this species. These "unique" outlier regions contained candidate genes for morphology, stress response and colouration, which both support previous research in this species and provide avenues for future investigation. Together, our results suggest that examining differentiation landscapes in the context of population history is a useful way for determining how natural selection shapes genomic variation in the wild.

Introduction

Since the first studies of allozyme variation in humans [1] and *Drosophila* [2,3], there has been great interest in explaining how evolutionary and ecological processes shape the patterns of genetic variation observed within and among natural populations. One focus of research and debate in this area has been on quantifying the roles of adaptive and neutral processes in explaining observed levels of genetic variation [4]. However, adaptation does not occur in isolation, but against a background of recombination, gene flow, and genetic drift. More recently there has been increased effort in understanding how these fundamental evolutionary forces operate in concert to generate and maintain the levels of genetic diversity commonly observed in natural populations [5,6].

The arrival of high-throughput sequencing and subsequent characterisation of genome-wide variation across large numbers of individuals has revealed that at the genetic level, patterns of variation and divergence among natural populations and species are highly heterogeneous [7]. A feature of these "genomic landscapes"

of divergence that has received particular attention is the presence of so-called "islands of differentiation"; outlier regions of the genome with high levels of divergence estimated from statistics such as F_{ST} or d_{xy} [7–10]. Initially these regions were termed "islands of speciation", and were thought to arise as a result of reduced gene flow in genomic regions associated with reproductive isolation [7,11]. Subsequent modelling and analysis has revealed that highly heterogeneous patterns of genomic divergence can occur in the complete absence of gene flow, as a result of recombination rate variation and linked selection [12,13]. Specifically, in genomic regions of low recombination, selection for beneficial mutations (positive selection), or against deleterious mutations (background selection), will impact relatively large genomic regions as a result of high levels of linkage disequilibrium (LD) among sites. Such selection at these regions reduces diversity within populations, and increases differentiation among them, resulting in "islands" of increased differentation that persist over evolutionary time [**???**,13]

Comparing patterns of genomic differentiation among sets of populations and/or species with different different levels of genome-wide divergence is a powerful way of disentangling the forces that shape variation among populations. Martin et al. [14] showed that, across multiple *Heliconius* butterfly populations and species, patterns of genomic variation were shaped by a combination of gene flow and selection, particularly in genomic regions harbouring genes involved in wing patterning. In contrast, Renaut et al. [9] showed that in *Helianthus* sunflowers, genomic architecture was the main driver of genomic differentiation across sets of populations. Similarly, recent research in birds has revealed that differentiation landscapes are conserved across populations, species and even across avian families [15–17]. This latter pattern appears to have arisen, at least in part, as a result of a highly conserved recombination landscape in birds.

It is now clear that the recombination landscape is a key driver of genomic variation within and among populations. However, we are only just beginning to understand how this recombination rate variation interacts with other evolutionary forces to shape patterns of differentiation across natural populations and species [18–22]. A recent, large-scale analysis in sticklebacks showed that, like in many other studies, islands of differentiation were most likely to be found in regions of low recombination, but by looking at this relationship across many pairs of populations it was shown that outlier genomic regions were more likely to arise in low recombination regions when gene flow occurred between populations [23]. There is also a significant impact of how recently separated populations are, and it is expected that in very recently separated populations the differentiation landscape is most likely to reflect selective sweeps, then as divergence accumulates, genomic architecture plays an increasingly important role [18]. More studies of genomic differentiation across very recently separated populations are now required to examine i) how rapidly divergence can accumulate in low recombining regions, and ii) how selection, gene flow and genomic architecture interact to shape genomic

divergence in the early stages of the differentiation process.

The European great tit (*Parus major major*) is an excellent model for ecological and evolutionary study [24]. Phyleogeographic research using mitochondrial and microsatellite loci suggests that this species has recently expanded across its European range, possibly from a single refugium in the Balkans [25]. Contemporary populations are characterised by large effective population sizes and high levels of gene flow among populations, resulting in low levels of genetic differentiation [26,27]. The genome of the great tit has recently been sequenced [28], and a high density (650K) panel of SNP markers has been developed. A recent study of two European populations using this marker panel suggests that rapid adaptation has occurred at the genomic and phenotypic levels [29]. A wealth of ecological data also exists across multiple great tit populations [30–32], enabling us to generate informed hypotheses about selection in this system.

In this study we examined how genomic architecture, natural selection and population history have shaped patterns of genomic variation across recently colonised European great tit populations. Using a large SNP panel typed across 28 populations (Fig. 1), we first characterise genome-wide patterns of variation within and among populations, in order to infer population history. We then examine how variation is partitioned across the genome, and test the hypothesis that highly divergent genomic regions have arisen in regions of low recombination. Finally, we examine how genomic divergence accumulates along the colonisation route of this species, with the aim of inferring how recent natural selection and background selection drive variation across the genome in the wild.

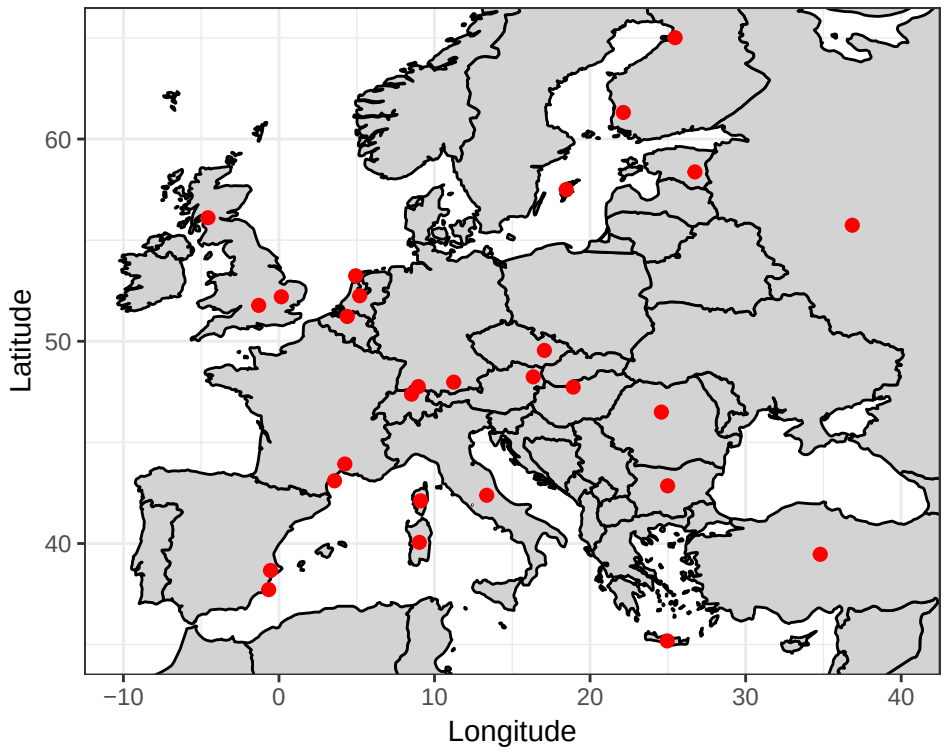

Figure 1 Sampling locations of great tit populations used in the present study.

Results

Genetic diversity and population history

Sampling locations and sample sizes for each population are given in Table 1. Levels of genetic diversity were generally high and LD declined rapidly in all populations, but we observed substantial differences in genetic diversity among populations (Fig. 2A). Highest levels of LD were observed in the island populations of Crete (Greece), Sardinia (Italy), Corsica (France) and Vlieland (Netherlands) with lowest levels of LD in central Europe, Turkey and Spain (Fig. S1). Similarly, heterozygosity was highest in central Europe, and lowest in Crete, Sardinia and the Balkans (Fig. 2A).

Considering all European great tit populations, there was no significant pattern of isolation-by-distance (Mantel test; r = 0.14, p = 0.16). Instead, the highest levels of F_{ST} were found in comparisons involving the island populations of Corsica, Sardinia and Crete (Fig. 2B). Considering only comparisons involving mainland populations, pairwise F_{ST} was still not related to related to geographic distance (r = 0.15, p = 0.13). We then examined genetic structure in relation to distance from potential refugial populations (only including mainland populations), and found that F_{ST} was significantly related to distance from Turkey (r =

0.84, $p < 0.001$; Fig. 2C) and the Balkans ($r = 0.47$, $p < 0.001$), but not to distance from Spain ($r = -0.07$, $p = 0.65$), or from Italy ($r = 0.15$, $p = 0.49$).

Admixture analysis revealed patterns of genetic structure corresponding to geographical variation in European great tits (Fig. S2). The $K = 2$ analysis assigned individuals in Sardinia and Corsica to one genetic cluster, and the remaining populations to the second. At $K = 3$ (the model that best fitted the great tit data; Fig. S3), Spain was separated from the rest of mainland Europe. Increasing values of K resulted of the separation of populations in Scotland ($K = 4$), Sardinia (from Corsica; $K = 5$), France ($K = 6$), Crete ($K = 7$) and England ($K = 8$). The Admixture output at $K = 8$ is displayed in Fig. 2C; further increases in K did not generate patterns of structure that corresponded to geographical variation (Fig. S2), and were increasingly less well supported (Fig. S3).

Maximum likelihood analyses implemented in TreeMix showed that a model with no migration explained 97.8% of variance in relatedness between populations; increasing the number of migration events substantially improved the percentage of relatedness explained, up to 99.7% when 10 migration events were fitted (Fig. S4). In Figure 3 we display the maximum likelihood tree with 2 migration events, after which the variance in relatedness explained plateaued when more migration events were added (Fig. S4). The tree was generally characterised by short branch lengths, with the exception of the island populations of Sardinia and Crete, which were grouped with the population from mainland Italy. The Spanish and Corsican populations were also grouped together, as were populations from Turkey and the Balkans, and populations from England and Scotland (Fig. 3). The two fitted migration edges both involved Sardinia, involving migration from eastern Europe, and to Corsica (Fig. 3).

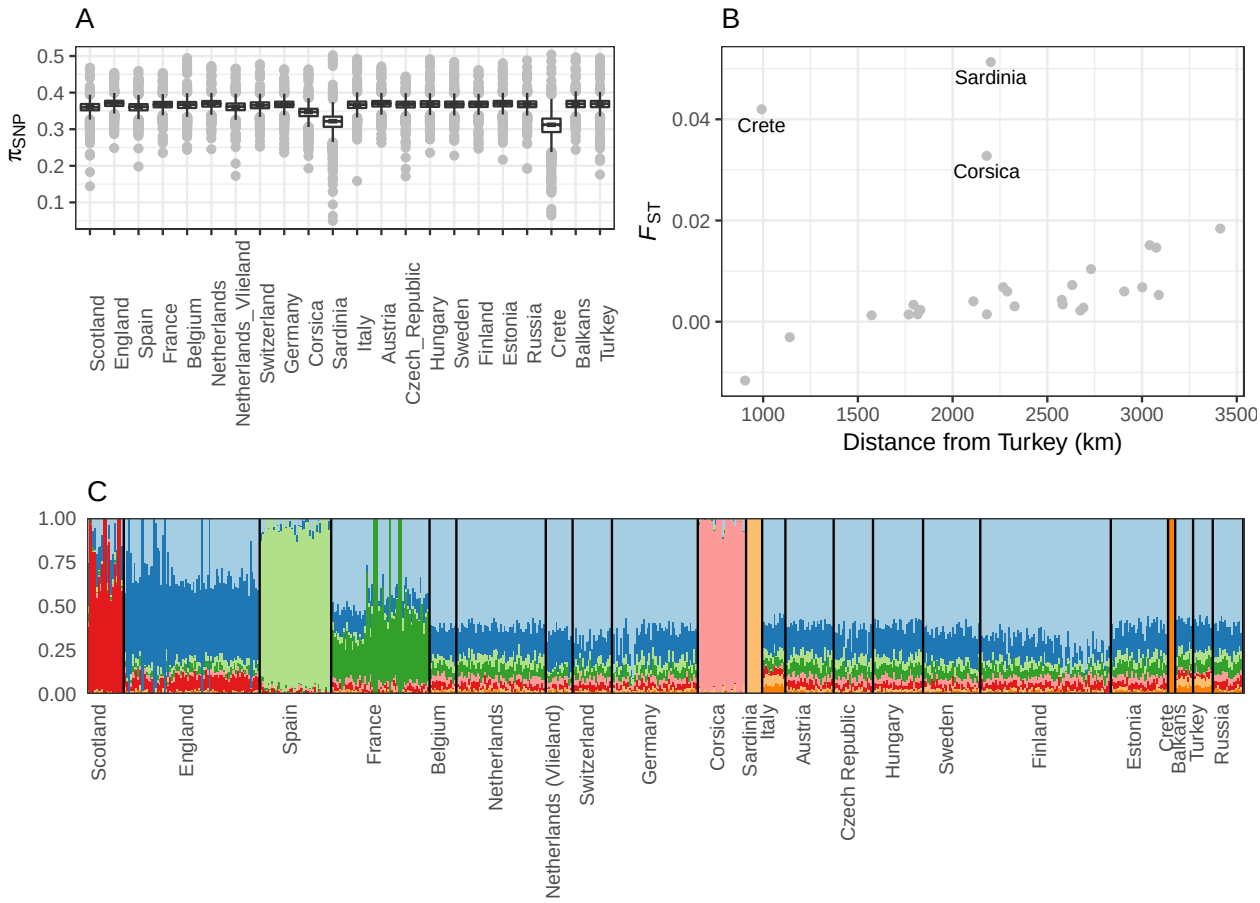

Figure 2 Genetic diversity and structure in European Great tit populations. **A** Nucleotide diversity within each population. **B** Pairwise F_{ST} in relation to geographic distance from the Turkey, only including comparisons involving Turkey. **C** Output from admixture analysis at $K = 8$.

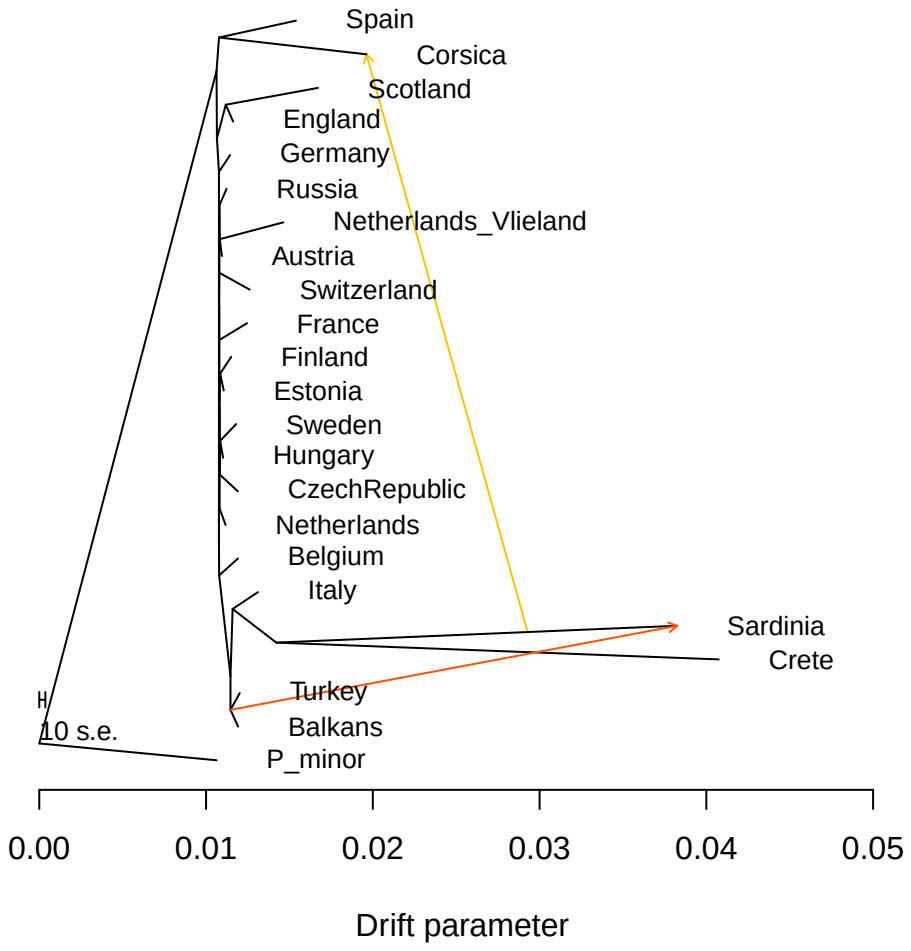

Figure 3 Maximum likelihood tree inferred by TreeMix, allowing ten migration events. Migrations events (arrows) are coloured according to their weight (red = higher migration), and horizontal branch lengths are proportional to the amount of genetic drift that has occurred along the branch.

Genomic landscapes of differentiation

To examine how landscapes of genomic divergence have formed along the colonisation route of European great tits, we calculated windowed F_{ST} in 500kb bins between each population and the proposed refugial populaiton in Turkey. We found that F_{ST} varied markedly across the genome in all comparisons (Fig. 4; Fig. S5). Outlier regions (windows with $zF_{ST} > 10$, where zF_{ST} is F_{ST} expressed relative to the genome-wide average, in standard deviations) were found in all comparisons apart from Crete and Sardinia, where overall levels of divergence were highest (Fig. 4). Of the 11 outlier regions, nine were found in only one or two comparisons, while the other two were found in 12 and 10 comparisons, respectively (Table S2). We hereafter refer to outlier regions found in one or two comparisons as "unique" outlier regions, and to those found in more than two comparisons as "shared" regions. Unique regions tended to be found in the most peripheral European great tits, with three found in Scotland, two in England, Spain and Finland; the remaining outlier regions were found in comparisons involving the Czech republic, Russia, Vlieland (Netherlands) and Belgium (Table S2).

Genes found in the outlier regions are displayed in Table S2. Perhaps most notable among these is COL4A5, a gene found to be associated with bill length, and under selection between populations between England and the Netherlands, in a recent great tit study [29]. Here we found that the COL4A5 region is an F_{ST} outlier in England and Scotland, but not in any other European populations (Table S2). Another notable candidate gene potentially involved in beak morphology, and previously found to be under selection in UK great tits is BMPR1A, which in this study was found in an outlier region in Scotland. Other genes in the unique outlier regions included potential candidates for morphology and obesity (PPP1CB [33], GHITM [34]), thermal stress response (CDKN1B [35]), malaria infection (MRPL33 [36]) and colour variation (SOX10 [37]).

Genomic differentiation with Turkey was negatively related to recombination rate in almost all populations (Fig. 5). The relationship between F_{ST} and recombination rare was generally weak, with correlation coefficients < 0.1. In a handful of populations this relationship was substantially stronger - most notably in the island populations of Corsica, Sardinia and Crete, and in England, Scotland and Spain (Fig. 5). Outlier regions of very high differentiation almost exclusively occurred in relatively low recombining regions, and accordingly, the recombination rate of outlier regions was significantly lower than the genome-wide average (Wilcoxon test, $P < 0.001$). There appeared to be no difference in the recombination rate between shared and unique outlier regions (Fig. 5), although the small number of regions precluded testing this hypothesis formally.

To further explore how selection may have shaped variation in F_{ST} outlier regions, we estimated levels

of nucleotide and haplotype diversity within these regions. Nucleotide diversity (π_{SNP}) in outlier regions varied from 0.28 to 0.41, and diversity in these regions was significantly lower than the genome-wide average (Wilcoxon test, $P = 0.007$; Fig. 6A). However, there appeared to be no difference in nucleotide diversity between shared and unique regions (Fig. 6A). Haplotype diversity varied substantially among regions, with haplotype richness ranging from 72 to 1033. Both haplotype richness and marker density in shared regions tended to be lower than those in unique regions (Fig. 6B). A detailed examination of haplotype structure in these regions is displayed in Figure S6. MIRTE TO EXPAND AND ADD FIGURE.

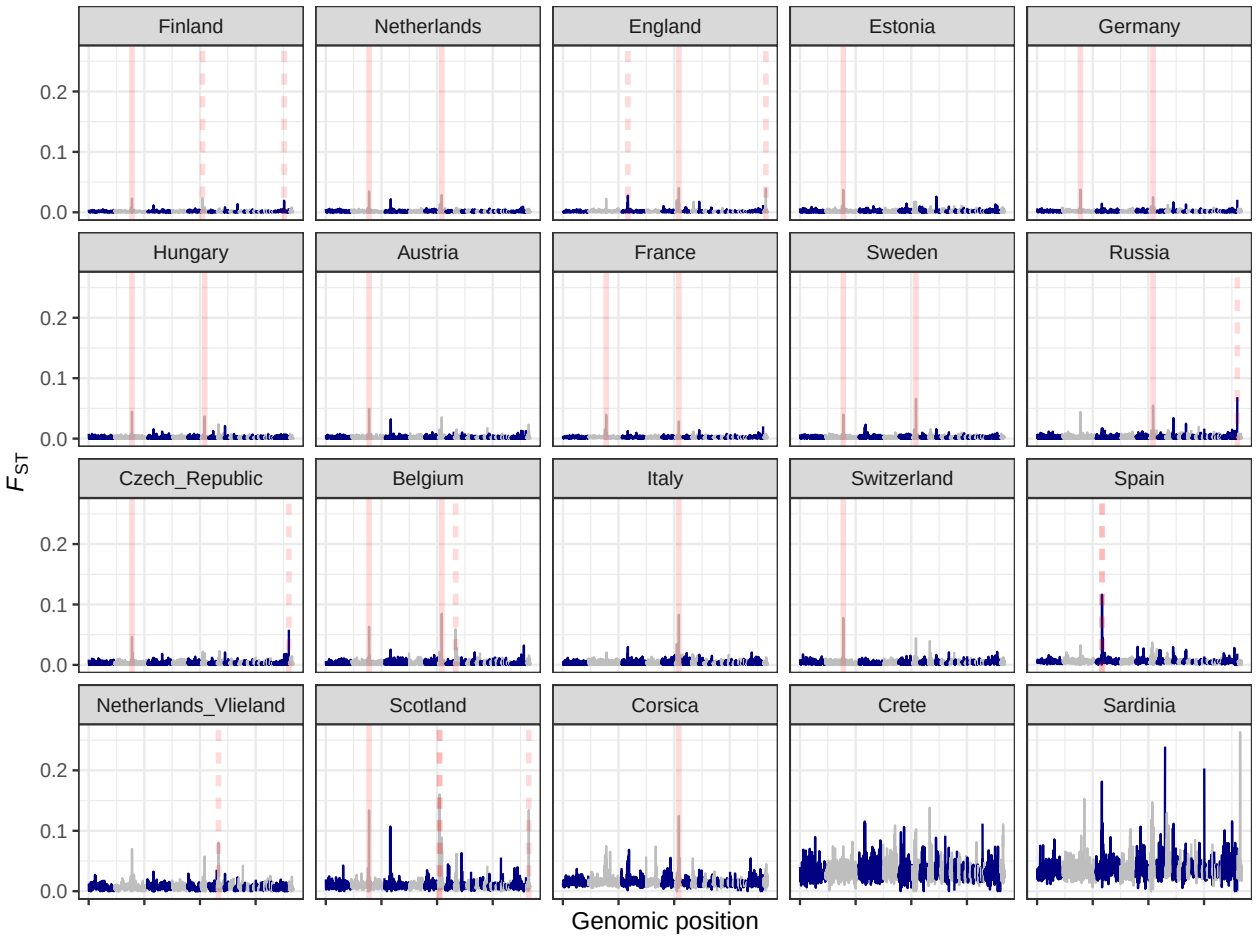

Figure 4 Landscapes of genomic differentiation in European great tit populations. F_{ST} across the genome is averaged in 500kb windows, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Red lines represent F_{ST} outliers (windows with mean F_{ST} values at least 10 standard deviations greater than the global mean for that comparison) shared across more than two comparisons (solid red lines), or unique to one or two comparisons (dashed red lines).

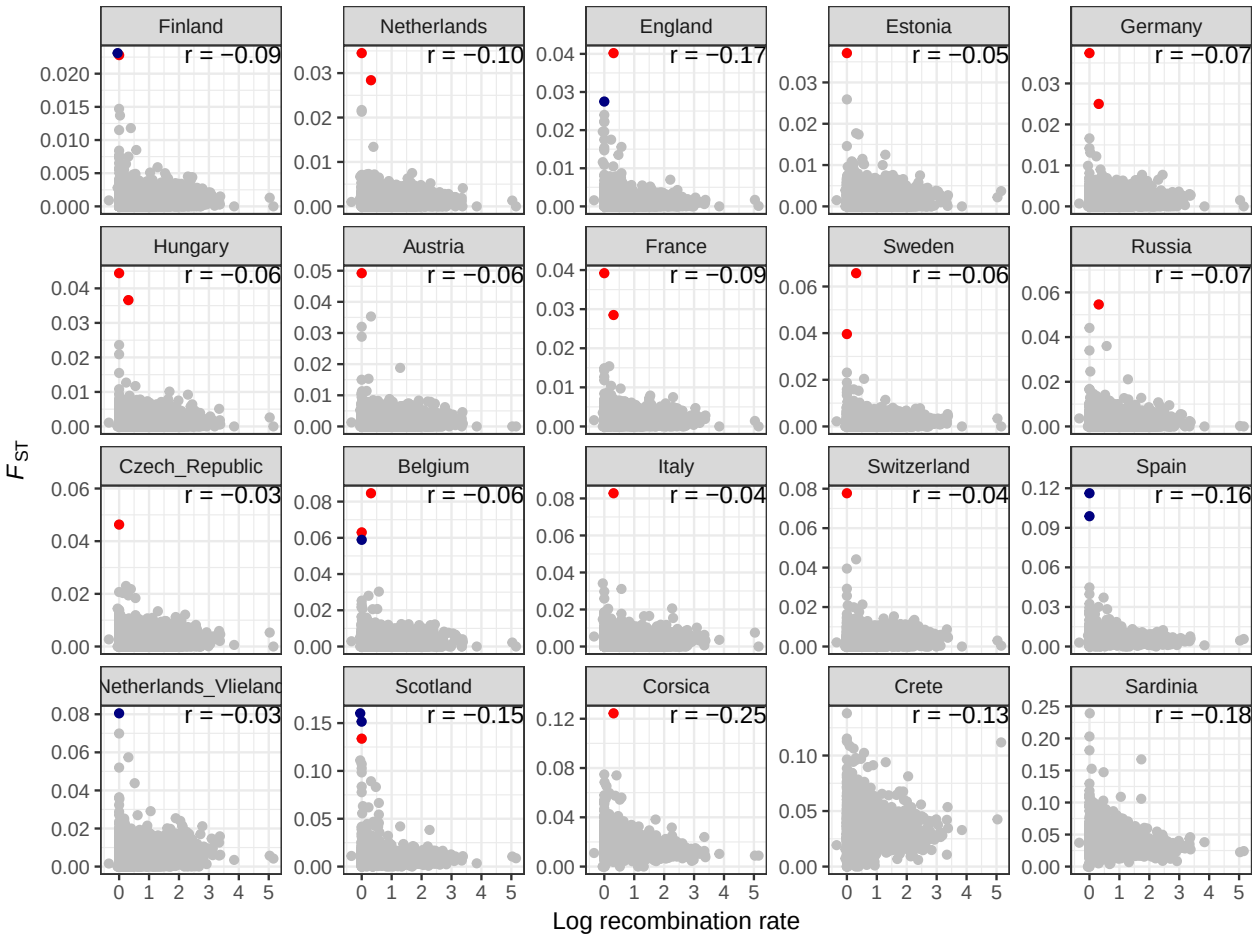

Figure 5 Genomic differentiation and recombination rate variation in European great tit populations. Each point is a 500kb window, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Coloured points represent F_{ST} outliers (mean zF_{ST} values > 10) shared across more than two comparisons (red), or unique to one or two comparisons (navy).

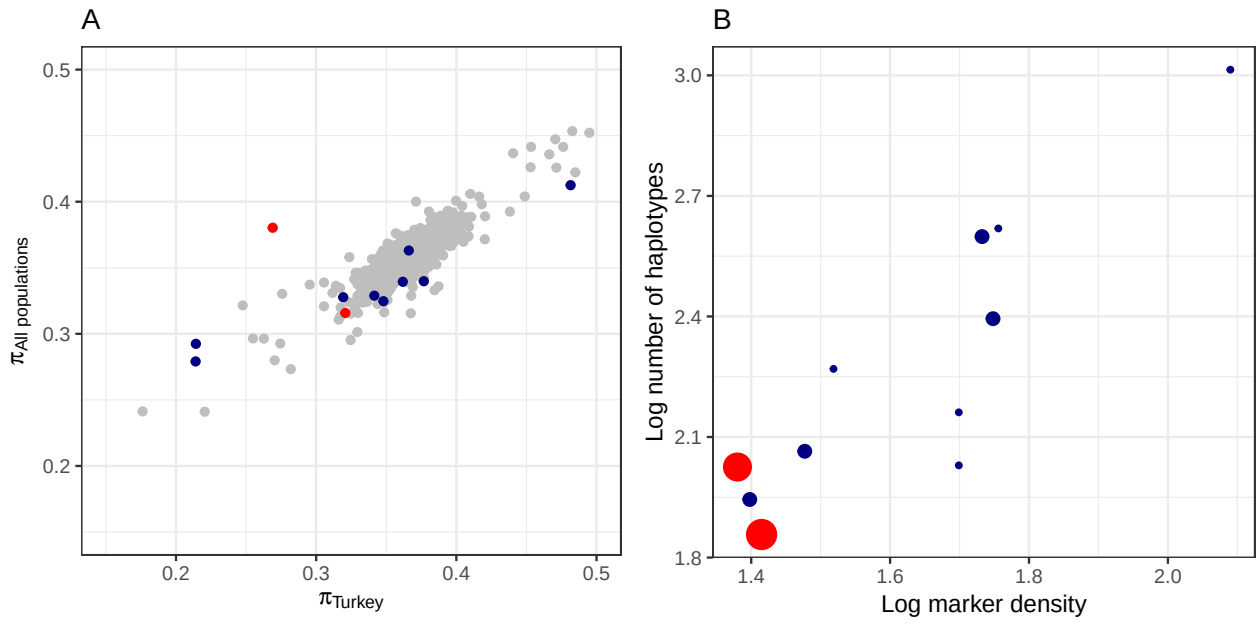

Figure 6 Genetic diversity in F_{ST} outlier regions in European great tit populations. Each point is an outlier window, with point size scaled by the number of populations the outlier was found in - "shared" haplotypes are coloured red and "unique" haplotypes navy (see main text for details). In **A**, Nucleotide diversity for each window is calculated for the proposed refugial population in Turkey, and plotted against the mean for all other populations. In **B**, SNP marker density is plotted against haplotype richness for each region.

Discussion

In this study we use a large panel of genetic markers to provide insight into the evolutionary history of the great tit in Europe. We found that, consistent with previous research in this species, levels of genomic variation were consistent with a large effective population size and gene flow among populations, and recent colonisation from a single refugial population in the east [26,28]. We then examined how genomic landscapes of differentiation vary along the proposed colonisation route of this species, and found evidence for both shared and unique islands of differentiation. In all cases, islands of differentiation arose in regions of low recombination and low diversity, but we found evidence for differences in haplotype structure and diversity among unique and shared outlier regions.

Our results suggest that great tits colonised Europe from a single refugium in Turkey and/or the Balkans. This scenario has been suggested before [25], and our observed pattern of increasing genome-wide divergence from the Turkish and Balkan populations lends this hypothesis empirical support. Genomic structure among mainland populations is, however, extremely low, and even with hundreds of thousands of markers Admixture

was unable to separate the central European populations. Similarly, in the TreeMix analysis the central European great tit populations formed a large group characterised by shallow branching, although much of the grouping that did occur (e.g. Finland and Estonia) made geographical sense. Interestingly, TreeMix grouped the Spanish and Corsican populations, which is consistent with previous subspecies descriptions of European great tits [38]. More generally, the highest levels of differentiation were found between island populations, suggesting a reduced effective population size and reduced gene flow in these populations, either as result of the colonisation process, or more recent bottlenecks [39].

It is likely that the many, and perhaps the majority, of wild populations are characterised by highly heterogeneous patterns of differentiation across the genome [22]. What is not yet clear is how quickly islands of differentiation arise, nor how selection, gene flow and drift interact with genomic architecture to shape differentiation over shorter and longer evolutionary time-scales [18]. Our results suggest that islands of differentiation arise in regions of low recombination even among populations that could be considered panmictic. European great tit populations have a mean F_{ST} of 0.007, yet islands of differentiation can clearly be seen, even among the most closely related population (Fig. 4, Fig. S5). Further, these islands were found in regions of low recombination and low diversity (Fig. 5, Fig. 6), suggesting that linked selection in regions of low recombination plays a key role in driving patterns of genomic differentiation even in the very earliest stages of population separation.

It is difficult to differentiate between positive and background selection in producing heterogeneous patterns of differentiation across the genome. Correlated patterns of genomic differentiation among independent sets of populations suggests that background selection likely plays a role [15–17]. It appears that European great tits have colonised from a single refugial population, and as such truly independent comparisons are not available. Therefore an outlier region shared among multiple populations could represent either selection in the ancestral population, or background selection. Given the pervasiveness of background selection in birds [16,17], and other organisms [40], and the overall negative relationship between F_{ST} and recombination rate in this study (Fig. 5), it is likely that this background selection is the driver of the shared outlier regions in great tits. Thus, it appears that background selection can generate islands of differentiation in the very earliest stages of population separation. This is not necessarily what we would expect - it is often assumed that peaks of high differentiation in recently separated populations are the product of selective sweeps, and that only after time do correlated patterns of genomic divergence arise [18,23]. Further research into genomic landscapes of differentiation among widespread continental species will shed further light on the role of background selection in shaping genomic divergence over short evolutionary timescales.

Regions of high differentiation that are not shared among populations are more likely to be the result of recent

positive selection [18]. We found evidence for such non-shared regions in European great tits, and that these tended to occur in peripheral populations. Observational and experimental research shows that adaptation at range edges is a key feature shaping divergence among recently colonised and expanding populations [41–43]. In great tits, previous research has shown evidence for adaptation in marginal populations. A comparison of Swedish and Finnish populations revealed genetic differences in basal metabolic rate, most likely an adaptive response to winter conditions [44]. Here we found outlier regions unique to populations in cold populations in Scotland, Finland and Russia, containing at least one candidate gene for thermal stress [35]. The detection of an outlier region containing the gene COL4A5 was particularly interesting, as this region was detected in a previous comparison of UK and Dutch great tits [29], in which significant associations with bill length and fitness were detected. Our results here are in accordance with the description of UK great tits as a separate subspecies [45], and combined with previous results suggest that this divergence is the result of recent natural selection in the UK [29]. Numerous other potential candidate genes were detected in these marginal populations, with potential roles in morphology, colouration, obesity and malaria infection. We are reluctant to speculate whether these candidate genes are genuine targets for natural selection, and more reluctant still to speculate as to how selection might be driving variation at these region. Regardless, these candidates will provide useful starting points for future genomic and ecological investigation.

Our data suggest that examining patterns of nucleotide and haplotype diversity in outlier regions may help to separate recent episodes of positive selection from the gradual process of background selection (Figs 6, S6). MIRTE TO EXPAND HERE. However, as we only detected a small number of outlier regions we are unable to make a detailed comparison between shared and unique regions. Further, simulations suggest that haplotype-frequency methods for detecting selection can be sensitive to background selection [46]. Research which combines multiple populations to allow shared and unique outlier regions to be identified, alongside a range of haplotype and SNP based methods for detecting selection, may allow us to identify the roles of positive and background selection in shaping variation.

Finally, it is worth noting that environmental association methods are highly suited method for detecting adaptation in widespread continental species [47,48], and further work will test how variation in the abiotic environment has shaped patterns of genomic variation in great tits. To our knowledge there has been little research on the potential for background selection to generate false positives in environmental association analyses, and given the pervasiveness of background selection, this should now be a priority. However, careful association of environmental and genotype data in species such as great tits, in which a wealth of ecological and genomic resources are available, is likely to generate interesting insights into the the genetic and phenotypic basis of natural selection.

Materials and Methods

Sampling and molecular methods

Samples were collected from 28 populations from 22 countries across Europe (Fig. 1; Table S1). Birds were trapped from nest boxes, or using mist nets, and ringed with a uniquely numbered aluminium ring. Blood was taken via brachial venipuncture, and stored in either 1 ml Cell Lysis Solution (Gentra Puregene Kit, Qiagen, USA) or Queen's buffer. All samples were genotyped using a custom-made Affymetrix® great tit 650K SNP chip at Edinburgh Genomics (Edinburgh, United Kingdom). After filtering, a total of 647 samples typed at 483888 SNPs were retained for analysis.

Analyses

Unless stated otherwise, all population genetic statistics were calculated in PLINK version 1.9 [49], and downstream analysis and plotting was carried out in R version 3.3 [50]. In each population, we estimated linkage disequilibrium (R^2) for each pair of markers within 50kb on the same chromosome, and compared this to physical distance between marker pairs. For a simple comparison of short-range LD among populations, we also calculated, for each population, mean LD between all pairs of markers within 1kb. We calculated observed heterozygosity for each SNP and population using a reduced SNP dataset, which was pruned based on linkage disequilibrium to remove all markers with $R^2 > 0.1$, then thinned with a probability of retaining each variant of 0.25.

We calculated genome-wide (mean) F_{ST} between each pair of populations using the pruned and thinned dataset described above. Pairwise F_{ST} was compared to geographic distance between populations using Mantel tests, implemented in the Ecodist package in R [51]. We tested whether genetic structure was related to distance from candidate refugial populations (in the Balkans, Turkey, Spain and Italy), using Pearson correlations. We also estimated population structure using Admixture version 1.3, with default settings [52]. We varied values of K from one to ten; by which point increasing values of K provided no informative information about population structure (see results). Model support for each value of K was estimated by calculating 5-fold cross-validation error. Finally, we visualised the evolutionary history among European great populations by generating a maximum likelihood tree in TreeMix version 1.13 [53]. We rooted the tree using a sample of *P. minor* individuals sampled from Amur, Russia. We fitted models allowing for range of migration events (0-10), and used a window size of 500 SNPs [53]. To assess model fit, we calculated the proportion of variance in relatedness between populations explained by each model [53].

We examined the genomic landscape of differentiation across European great tit populations by calculating

F_{ST} in 500kb bins, using python scripts obtained from Github (https://github.com/simonhmartin/genomics_general). We did not estimate d_{xy}, as this parameter is difficult to estimate accurately from single SNP loci [13]. We also calculated standardised F_{ST} (hereafter ZF_{ST}) by mean-centring windowed values and dividing them by the standard deviation among windows. We defined outlier regions as 500kb bins with zF_{ST} values greater than ten. We tested whether the landscape of genomic differentiation was related to recombination rate variation using a recombination map previously developed for the great tit using a 10K SNP chip [54], from which we estimated recombination rates in 500kb non-overlapping windows.

To gain further insight into how selection may have shaped variation in outlier regions, we examined patterns of haplotype variation. MIRTE TO ADD.

Acknowledgements

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